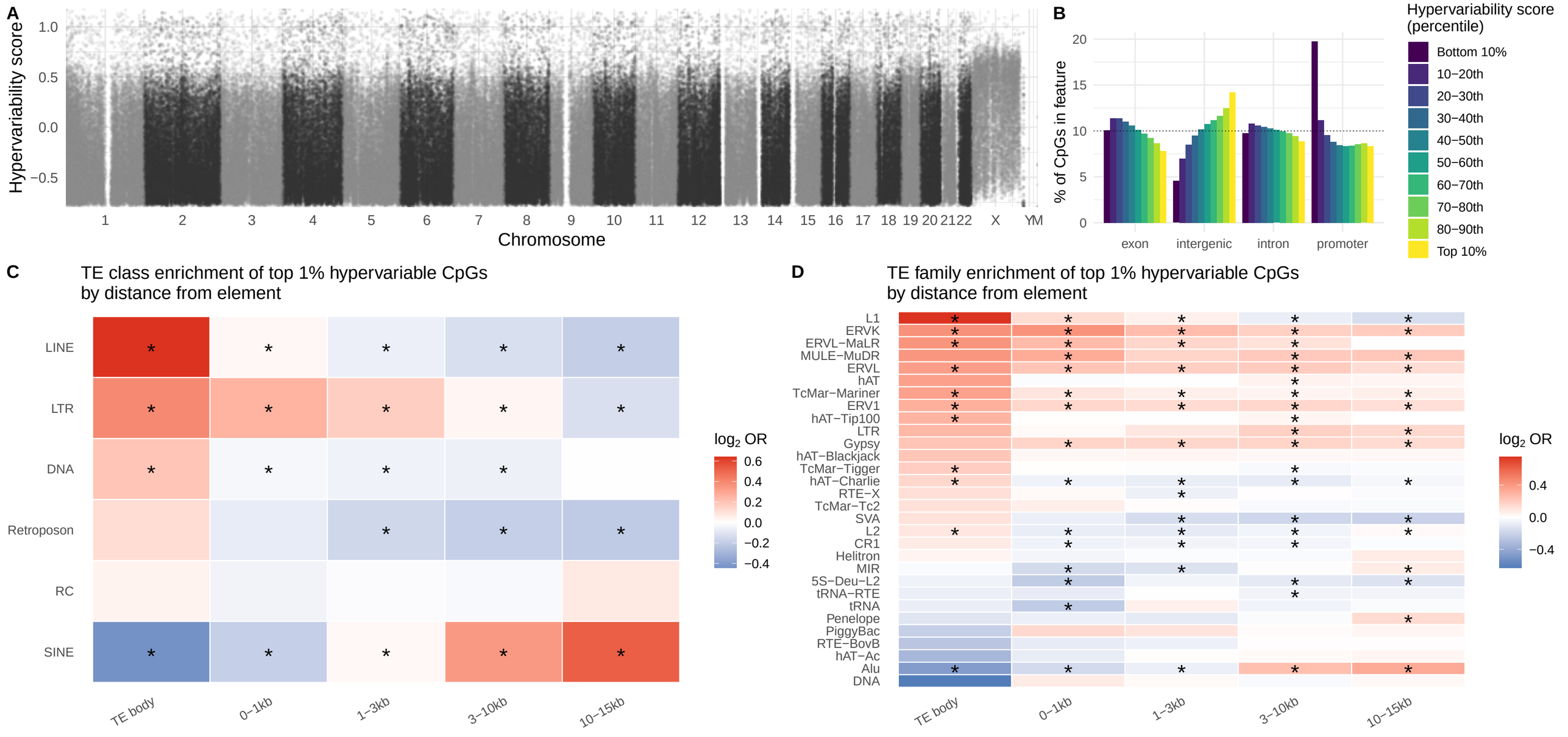

A
Hypervariability score
1.0
0.5
0.0
−0.5
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 X Y
Chromosome
B
% of CpGs in feature
20
15
10
5
0
exon
intergenic
intron
promoter
Hypervariability score (percentile)
Bottom 10%
10−20th
20−30th
30−40th
40−50th
50−60th
60−70th
70−80th
80−90th
Top 10%
C
TE class enrichment of top 1% hypervariable CpGs by distance from element
LINE
LTR
DNA
Retroposon
RC
SINE
TE body
0−1kb
1−3kb
3−10kb
10−15kb
log2 OR
0.6
0.4
0.2
0.0
−0.2
−0.4
D
TE family enrichment of top 1% hypervariable CpGs by distance from element
L1
ERVK
ERVL−MaLR
MULE−MuDR
ERVL
hAT
TcMar−Mariner
ERV1
hAT−Tip100
LTR
Gypsy
hAT−Blackjack
TcMar−Tigger
hAT−Charlie
RTE−X
TcMar−Tc2
SVA
L2
CR1
Helitron
MIR
5S−Deu−L2
tRNA−RTE
tRNA
Penelope
PiggyBac
RTE−BovB
hAT−Ac
Alu
DNA
TE body
0−1kb
1−3kb
3−10kb
10−15kb
log2 OR
0.4
0.0
−0.4
E
hvCpG enrichment at KZFP binding sites by distance from peak
Odds ratio (top1% hvCpG vs background)
1.0
0.9
0.8
peak
0−2kb
2−4kb
4−6kb
6−8kb
8−10kb
F
hvCpG enrichment around KZFP sites: footprint, TE body, and (non−TE) flanks
Odds ratio (top1% hvCpG vs background)
1.2
1.0
0.8
0.6
KZFP peak
TE body
0−2kb
2−4kb
4−6kb
6−8kb
8−10kb
n.s.
FDR < 0.05
KZFP in TE
KZFP not in TE
G
hvCpG enrichment: KZFP−bound vs unbound TEs
TE−conditional OR = 1.11 (FDR 2.8e−60)
TE: KZFP−bound
TE: unbound
1.00
1.05
1.10
1.15
Odds ratio (vs background)